HYDROTHERMAL SYNTHESIS OF THE SPINEL COMPOUND $CuCo_2S_4$

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Abstract

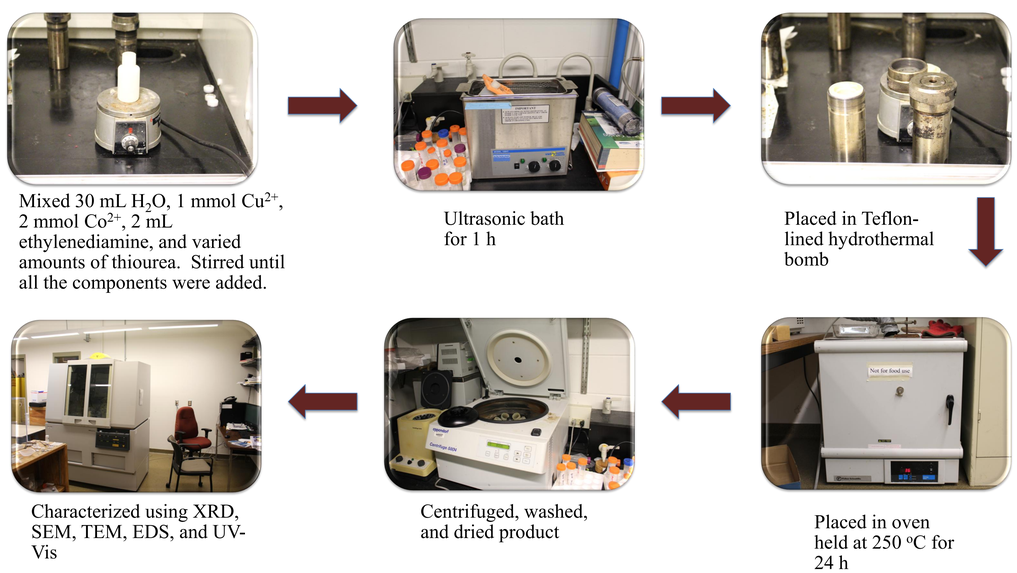
Transition metal sulfides have become the focus of intense research because they have potential for use as super capacitor electrodes, solar cell absorber layers, etc. Copper cobalt (II) sulfide ($CuCo_2S_4$), also known as carrollite, is a tertiary sulfide that occurs naturally. In this research, nanocrystalline $CuCo_2S_4$ was successfully synthesized using a hydrothermal approach, which is a less toxic and inexpensive method. The most important reaction variables were temperature, molar ratio of the metal cations to the sulfur source (thiourea), and presence of ethylenediamine in the reaction mixture. $CuCo_2S_4$ was synthesized using a slight excess of the sulfur source, which was thiourea, and heating hydrothermally at 250 ^oC for 24 h. The addition of ethylenediamine was necessary to produce a pure compound.

Introduction

The goal of the research is to synthetically prepare carrollite as nanocrystalline particles using:

- Less toxic and inexpensive reagents
- Hydrothermal route
  - High temperature
  - High pressure
- Synthetic variables:
  - Metal to sulfur ratio
  - Addition of ethylenediamine
  - Temperature (room temperature vs. 250 ° C)
  - Time
- Characterization Techniques
  - X-ray diffraction (XRD)
  - Scanning electron microscopy (SEM) with an energy-dispersive X-ray spectrometry (EDS)
  - Ultraviolet-visible spectroscopy (UV-Vis)
  - Transmission electron microscopy (TEM)

Experimental Approach



Results

Legend For XRD Data

"." – Cu_7S_4 PDF # 00-024-0058, "*" – Co_9S_8 PDF # 01-086-2273 Units: Intensity (Arbitrary Units) and 2 theta (Degrees)
"#" – CoS_2 PDF # 00-041-1471, "^" – Cu_9S_5 PDF # 00-023--0960
"!" – CoS PDF # 03-065-3418 $CuCo_2S_4$ PDF # 00-042-1450

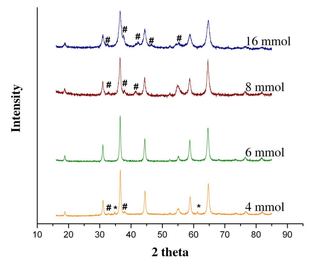


Fig. 1. XRD pattern of the samples prepared varying the concentration of thiourea

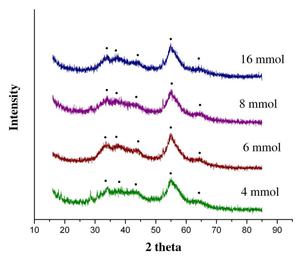


Fig. 2. XRD pattern of the samples prepared at room temperature while varying the concentration of thiourea

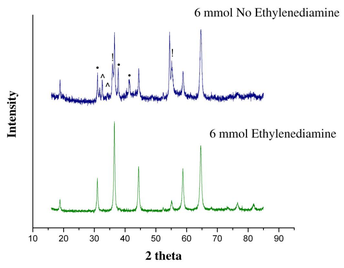


Fig. 3. XRD patterns of the 6 mmol thiourea sample prepared with and without ethylenediamine

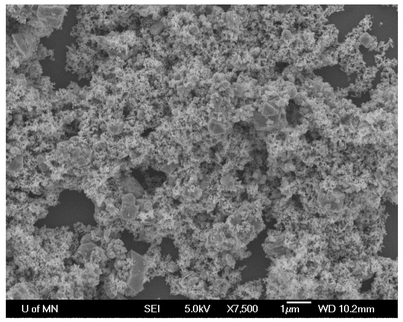


Fig. 4. SEM image of the sample prepared using 6 mmol of thiourea

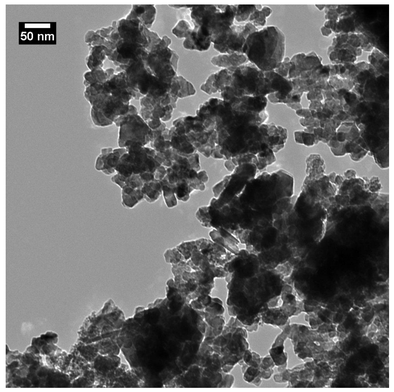
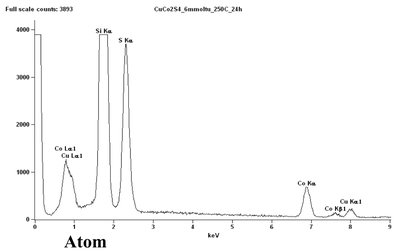


Fig. 5. TEM image of the sample prepared using 6 mmol of thiourea



Atom %

Spot	1	2	3	4	5	Avg	Comp.
Cu	14.2	14.4	14.36	15.29	13.33	14.32	1
Co	30.97	31.09	30.86	30.53	30.56	30.80	2.2
S	54.82	54.51	54.78	54.18	56.11	54.88	3.8
Total	100	100	100	100	100	100	

Fig. 6. Typical EDS analysis of the sample prepared using 6 mmol of thiourea

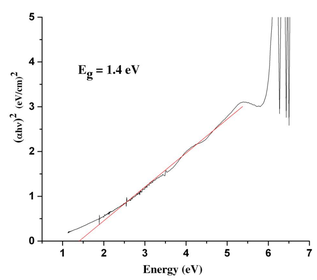


Fig. 7. Tauc plot that enables determination of the band gap using UV-Vis spectroscopy and applying the Tauc equation of the sample prepared using 6 mmol thiourea

Conclusion

- Crystalline $CuCo_2S_4$ was successfully synthesized at 250 ^oC using the hydrothermal approach
  - Ideal metal to sulfur ratio in the reaction mixture is 1:2
  - Average particle size ~25 nm
- Room temperature synthesis did not produce Crystalline $CuCo_2S_4$
- Addition of ethylenediamine improved purity of the product dramatically
- $CuCo_2S_4$ band gap experimentally determined to be 1.4 eV

Future Work

- Synthesizing additional compounds with different compositions: e.g., $CoNi_2S_4$, $NiCo_2S_4$, $MnCo_2S_4$
- Testing for the efficiency of the compounds as thin film depositions for solar cells